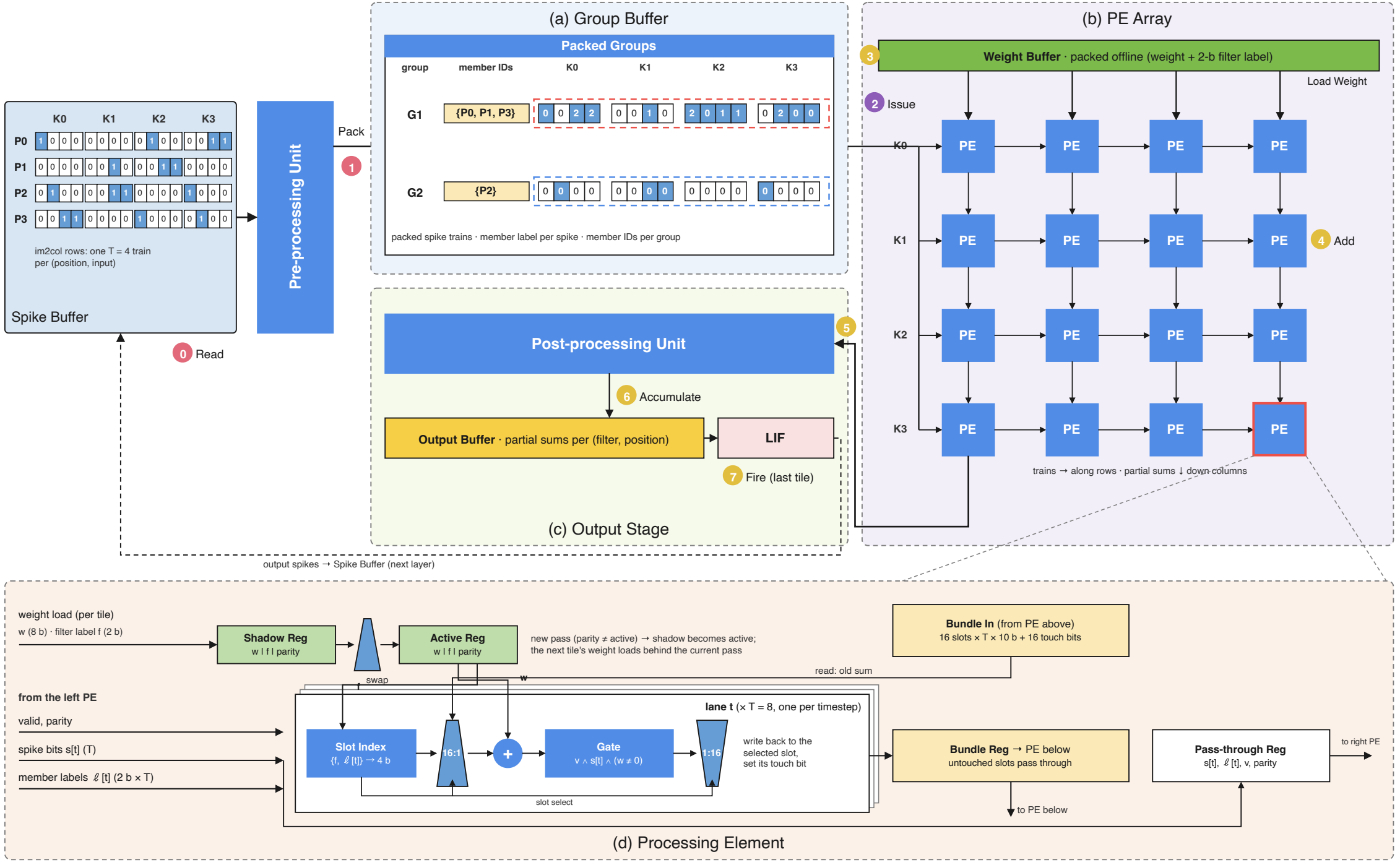

(a) Group Buffer
Packed Groups
group
member IDs
K0
K1
K2
K3
G1
{P0, P1, P3}
G2
{P2}
packed spike trains · member label per spike · member IDs per group
Pack
1
Spike Buffer
K0
K1
K2
K3
P0
P1
P2
P3
im2col rows: one T = 4 train per (position, input)
Pre-processing Unit
0 Read
(b) PE Array
3
Weight Buffer · packed offline (weight + 2-b filter label)
Load Weight
2 Issue
PE
4 Add
trains → along rows · partial sums ↓ down columns
(c) Output Stage
Post-processing Unit
5
6 Accumulate
Output Buffer · partial sums per (filter, position)
LIF
7 Fire (last tile)
output spikes → Spike Buffer (next layer)
(d) Processing Element
weight load (per tile)
w (8 b) · filter label f (2 b)
Shadow Reg
w l f l parity
Active Reg
w l f l parity
swap
new pass (parity ≠ active) → shadow becomes active;
the next tile's weight loads behind the current pass
Bundle In (from PE above)
16 slots × T × 10 b + 16 touch bits
read: old sum
from the left PE
valid, parity
spike bits s[t] (T)
member labels ℓ [t] (2 b × T)
lane t (× T = 8, one per timestep)
Slot Index
{f, ℓ [t]} → 4 b
16:1
Gate
v ∧ s[t] ∧ (w ≠ 0)
1:16
write back to the selected slot, set its touch bit
slot select
Bundle Reg → PE below
untouched slots pass through
to PE below
Pass-through Reg
s[t], ℓ [t], v, parity
to right PE